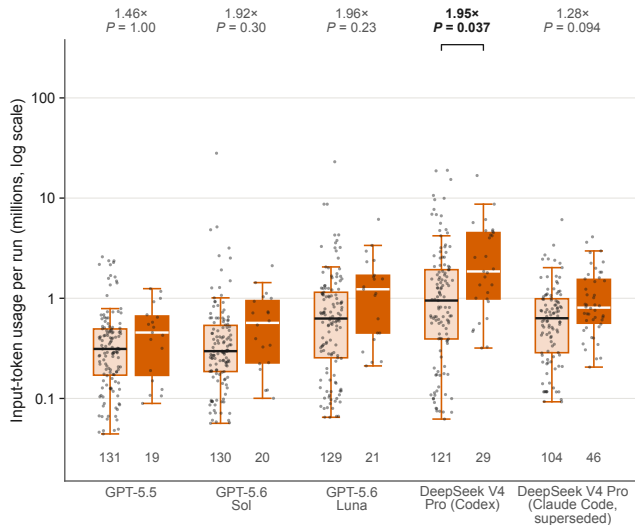


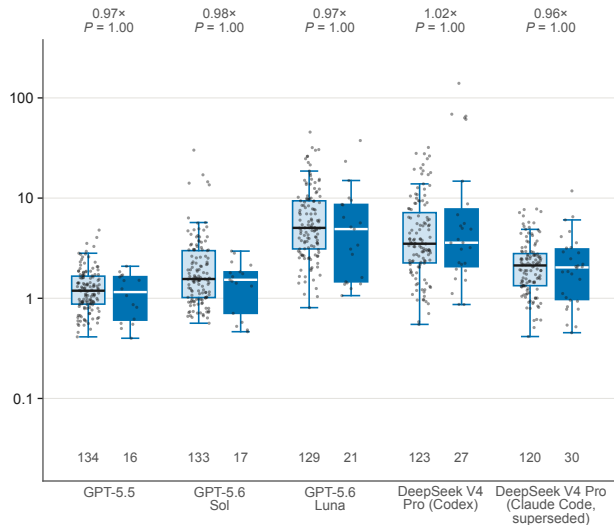
a Open-ended code condition: scored-incorrect runs used more input tokens, significantly so only for DeepSeek V4 Pro (Codex)

Same task and model configuration (33 split replicate sets): scored-incorrect runs used 0.99 times the input tokens of their scored-correct siblings (median; Wilcoxon signed-rank $P = 0.74$)



b Galaxy condition: no significant difference between scored-incorrect and scored-correct runs

Same task and model configuration (28 split replicate sets): scored-incorrect runs used 1.01 times the input tokens of their scored-correct siblings (median; Wilcoxon signed-rank $P = 0.96$)



▮ Open-ended code, scored correct
 ▮ Open-ended code, scored incorrect
 ▮ Galaxy, scored correct
 ▮ Galaxy, scored incorrect

BixBench-Verified-50; every run is a dot (1,500 runs; numbers along the bottom give the runs per box). Boxes span the middle 50% of runs and mark the median; whiskers extend to 1.5 times the interquartile range. Above each pair: median input-token usage, scored incorrect + scored correct, and the two-sided Mann–Whitney U P value, Holm-adjusted over the ten comparisons (bold with a bracket: $P < 0.05$). The three replicate runs of a task are not independent, so these P values may be optimistic. Source Data give each ratio's 95% cluster-bootstrap interval (source capsules, 20,000 resamples); for DeepSeek V4 Pro (Codex) it is 0.99 to 6.41, so this difference does not survive that dependence. Input-token usage includes cached input; output tokens (0.6% of input) are not added. Scored-incorrect runs include runs that submitted no answer.